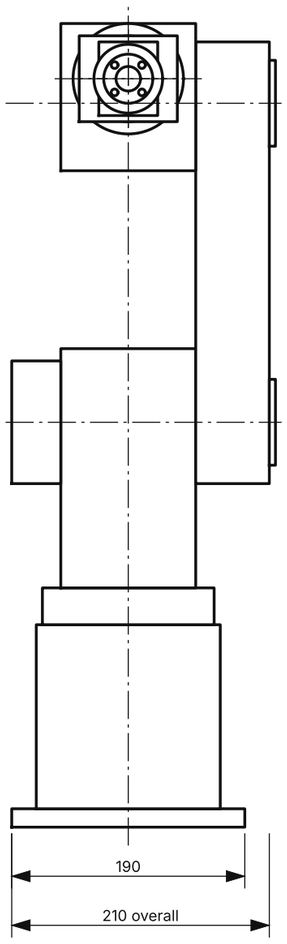
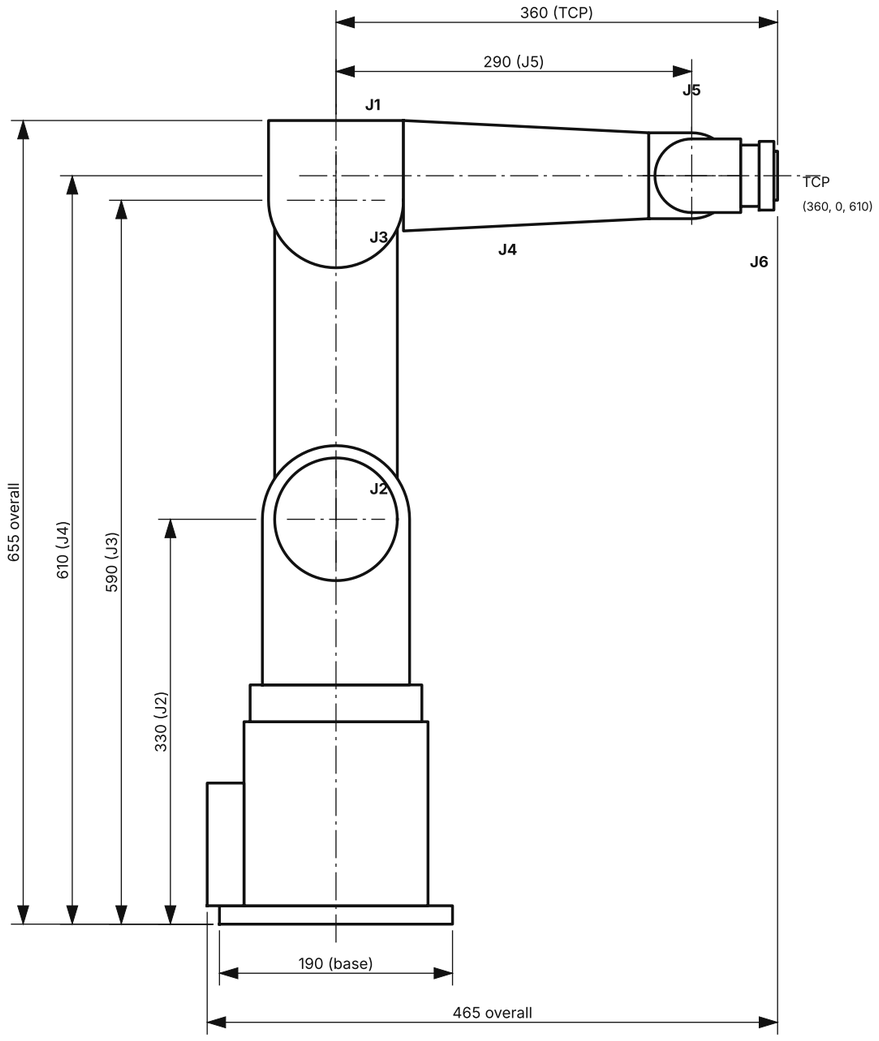
RIGHT SIDE VIEW (YZ)

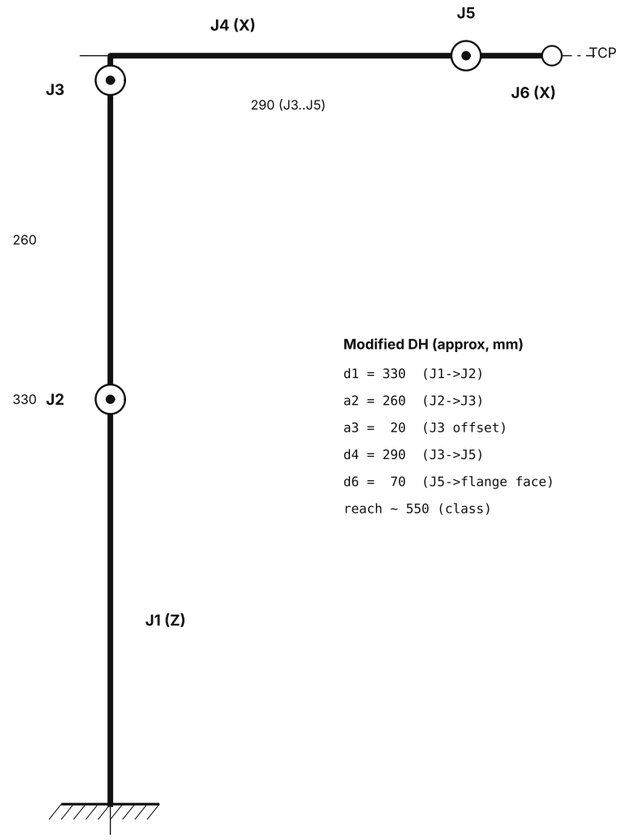
(1st angle: placed left of front view)



FRONT VIEW (XZ)



KINEMATIC SCHEMATIC (home pose, XZ plane)



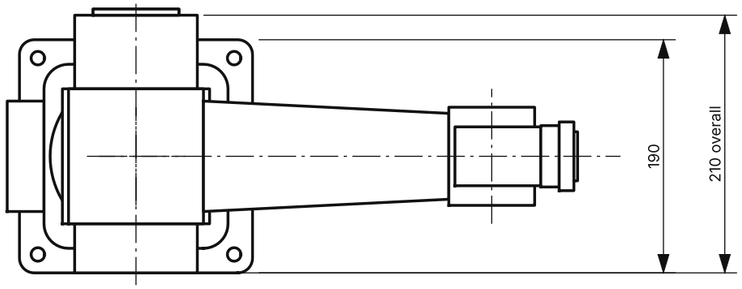
Modified DH (approx, mm)

```
d1 = 330  (J1->J2)
a2 = 260  (J2->J3)
a3 =  20  (J3 offset)
d4 = 290  (J3->J5)
d6 =  70  (J5->flange face)
reach ~ 550 (class)
```

NOTES

1. Home pose: all joints at 0 deg.
2. World: Z up = J1 axis, +X = arm reach,
   J2/J3/J5 parallel to Y, J4/J6 along X.
3. Hub caps / fillets omitted (cosmetic).
4. Brand marks intentionally not shown.

TOP VIEW (XY)



JOINT TABLE

Axis	Between	Axis dir.	Axis passes (x, z)	Limits [deg]
J1	Base - Turret	Z	(0, 0)	-170 / +170
J2	Turret - Lower arm	Y	(0, 330)	-110 / +120
J3	Lower arm - Elbow	Y	(0, 590)	-70 / +200
J4	Elbow - Forearm	X	(-, 610)	-190 / +190
J5	Forearm - Wrist	Y	(290, 610)	-125 / +125
J6	Wrist - Flange	X	(-, 610)	-360 / +360

Limits approximate (verify against official datasheet).

PARTS LIST

Item	Part (CATPart)	Home pos X,Y,Z (mm)
P01	P01_Base - Base pillar	0, 0, 0
P02	P02_Turret - Turret / shoulder	0, 0, 165
P03	P03_LowerArm - Lower arm	0, 85, 330
P04	P04_ElbowHousing - Elbow housing	0, 0, 590
P05	P05_Forearm - Forearm	55, 0, 610
P06	P06_Wrist - Wrist	290, 0, 610
P07	P07_Flange - Tool flange (ISO)	345, 0, 610

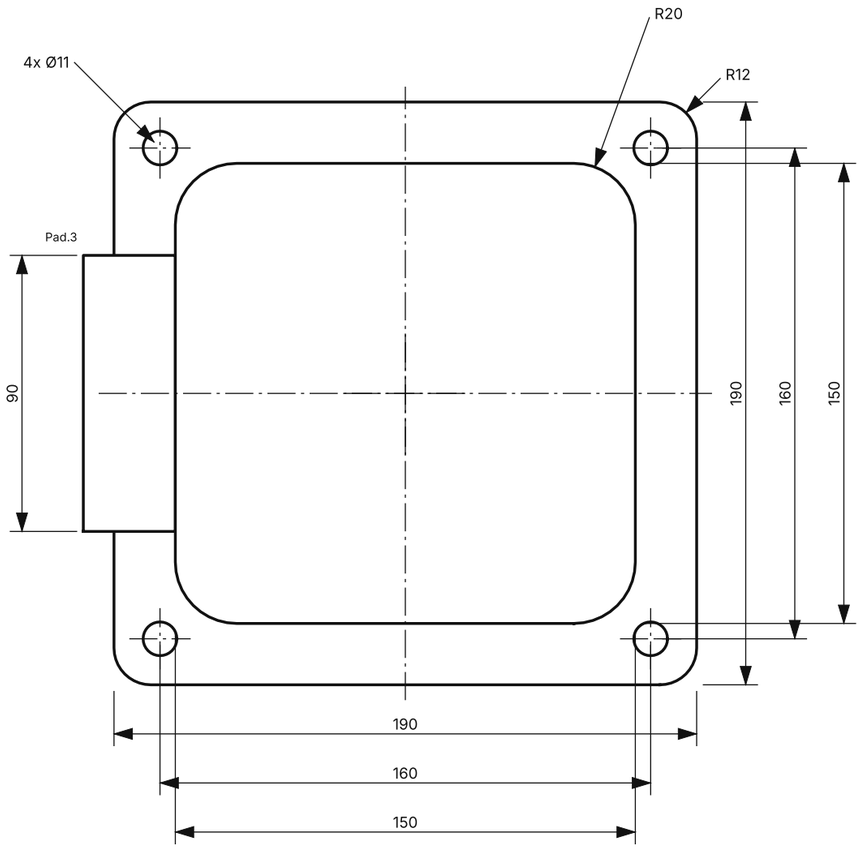
6-Axis Compact Robot Arm (LR Mate 200iD/4SC style) - approximate

SHEET 1 - GENERAL ARRANGEMENT (home pose)

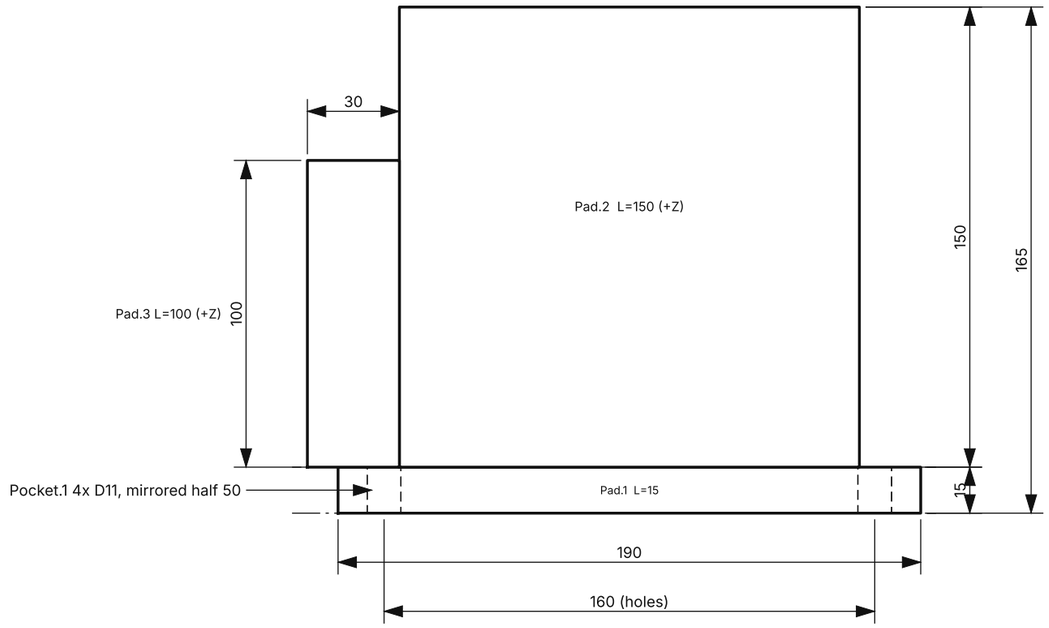
SCALE	UNITS	PROJECTION	SHEET	DATE
1:5	mm	1st angle	1/8	2026-11-11

Dimensions approximate - verify before manufacture

VIEW A - TOP (sketch plane XY: Sketch.1, .2 at z=0; Sketch.3, .4 at z=15)



VIEW B - FRONT (XZ): extrusion thickness



chain lines = sketch planes z=0 and z=15

CATIA FEATURE RECIPE (Part Design)

```
1 Sketch.1  plane z=0 (H=+X,V=+Y): rounded rect 190x190 R12      -> Pad.1  L=15 (+Z)    z 0..15
2 Sketch.2  plane z=0 (H=+X,V=+Y): 4x circle D11 at (+-80,+-80)  -> Pocket.1 Mirrored extent half 50
3 Sketch.3  plane z=15 (H=+X,V=+Y): rounded rect 150x150 R20     -> Pad.2  L=150 (+Z)   z 15..165
4 Sketch.4  plane z=15 (H=+X,V=+Y): rect X-105..-75, Y-45..45   -> Pad.3  L=100 (+Z)   z 15..115
Local frame = world frame, origin at floor centre. Part is FIXED in the assembly.
```

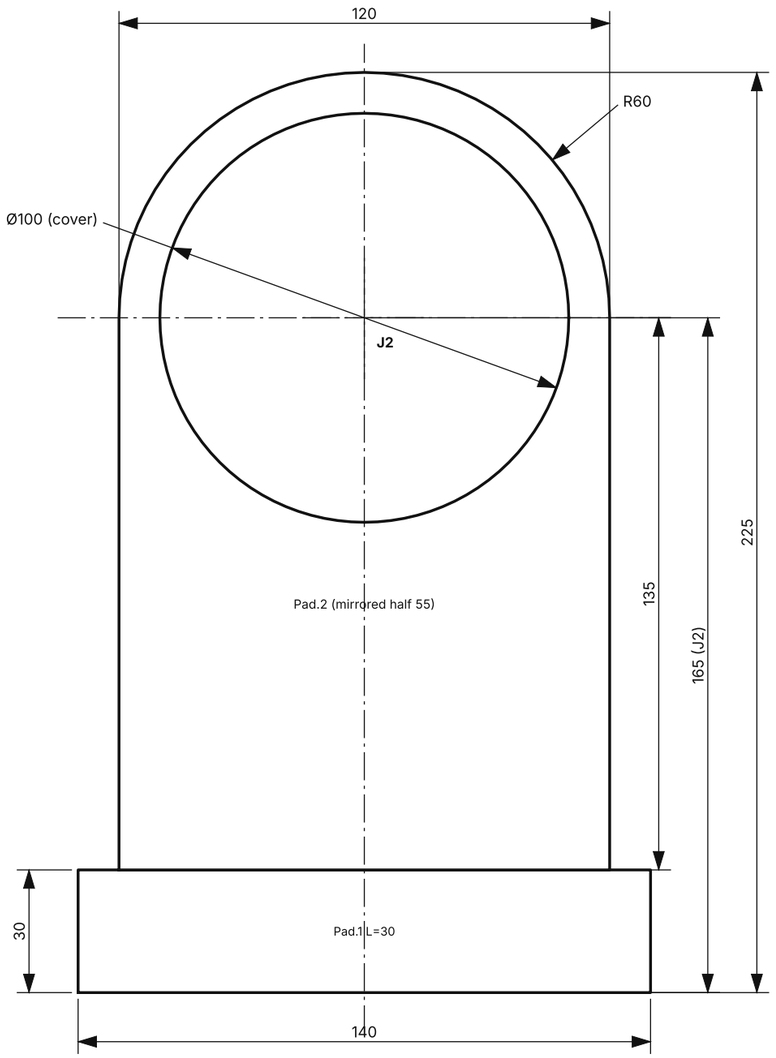
6-Axis Compact Robot Arm (LR Mate 200iD/4SC style) - approximate

P01_Base - sketch profiles and extrusion

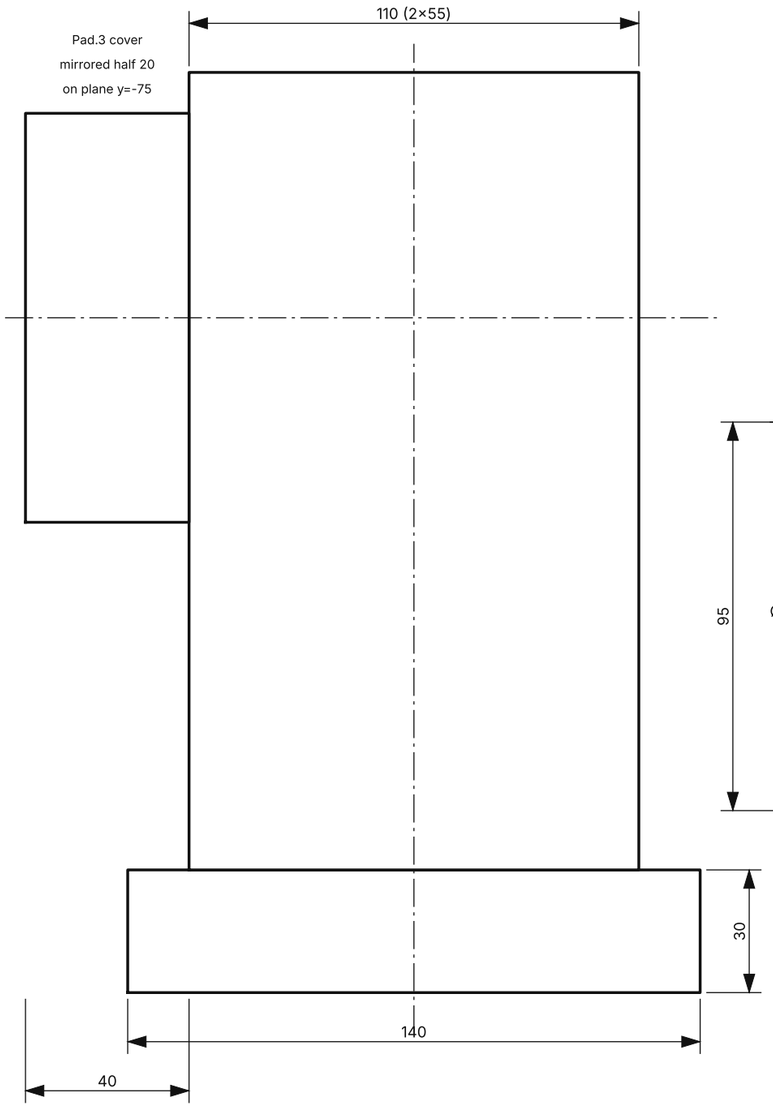
SCALE	UNITS	PROJECTION	SHEET	DATE
1:2	mm	1st angle	2/8	2026-11-11

Dimensions approximate - verify before manufacture

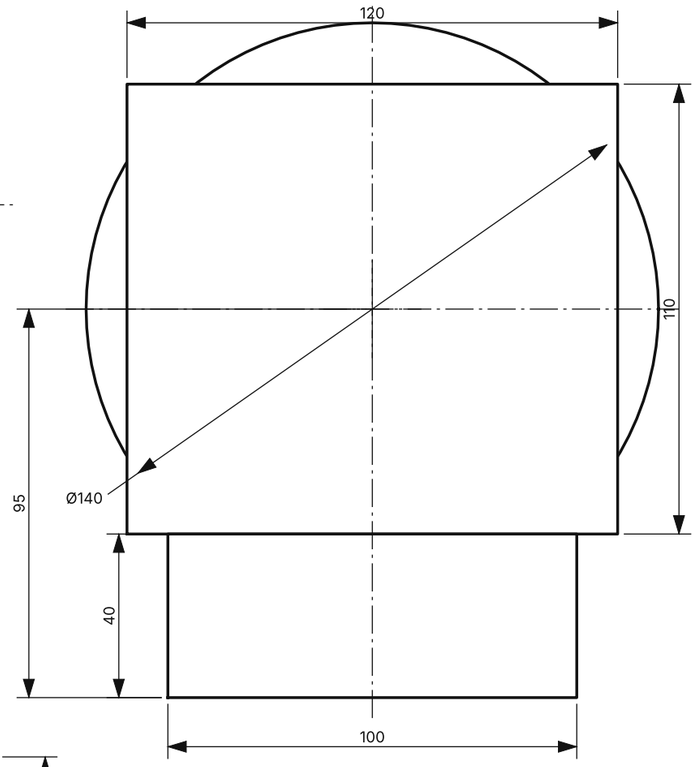
VIEW A - FRONT (XZ): Sketch.2 shoulder, Sketch.3 cover



VIEW B - SIDE (YZ)



VIEW C - TOP (XY)

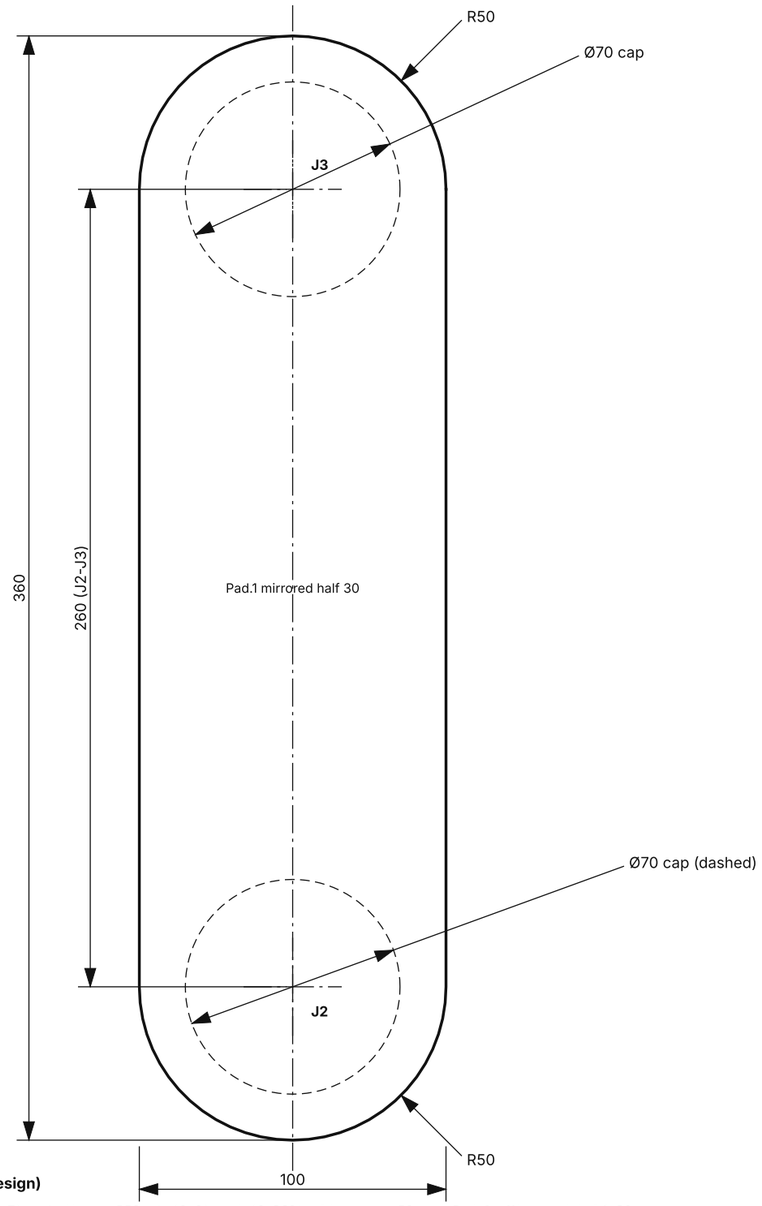


CATIA FEATURE RECIPE (Part Design)

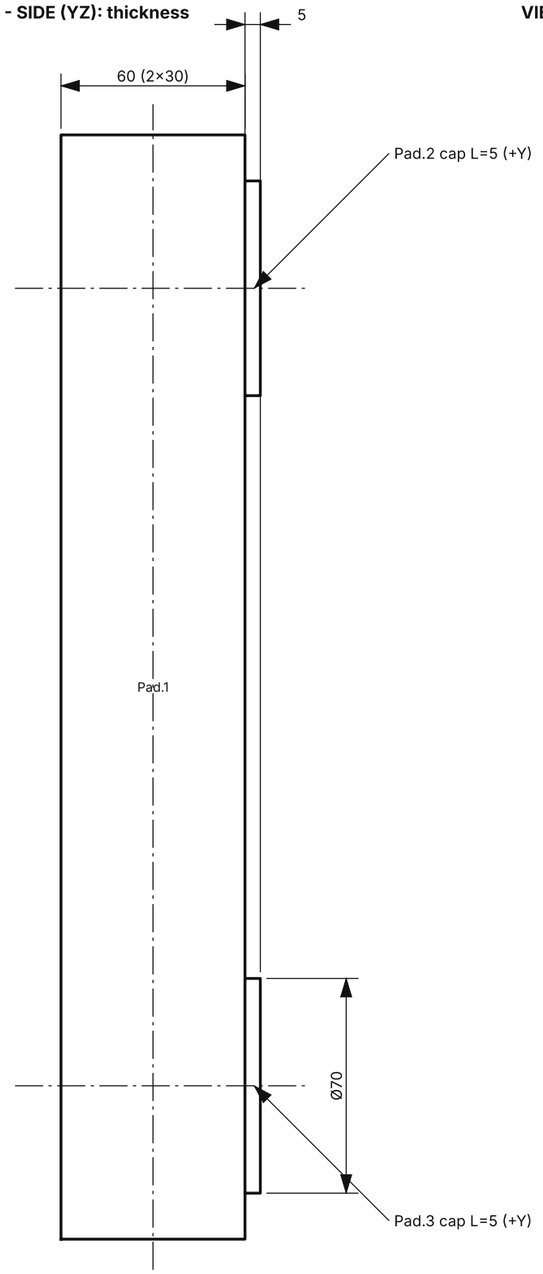
```
1 Sketch.1  plane z=0 (H=+X,V=+Y): circle D140 at (0,0)                -> Pad.1  L=30 (+Z)   z 0..30
2 Sketch.2  plane y=0 (H=+X,V=+Z): (-60,30)-(60,30)-(60,165), arc R60 c(0,165) over top to (-60,165), close
                                                                        -> Pad.2  Mirrored extent half 55 (Y +-55)
3 Sketch.3  plane y=-75 (H=+X,V=+Z): circle D100 at (0,165)            -> Pad.3  Mirrored extent half 20 (Y -95..-55)
Local origin = (0,0,165) world (top of base).  J2 axis (parallel Y) at local z=165 -> world z=330.
```

6-Axis Compact Robot Arm (LR Mate 200iD/4SC style) - approximate				
P02_Turret - sketch profiles and extrusion				
SCALE 1:1.5	UNITS mm	PROJECTION	SHEET 1st angle 3/8	DATE 2026-11-11
Dimensions approximate - verify before manufacture				

VIEW A - FRONT (XZ): Sketch.1 stadium



VIEW B - SIDE (YZ): thickness



VIEW C - hub cap sketch axes

Sketch.2 on plane y=30:

origin (0,30,0) H=(0,0,1) V=(1,0,0)

normal = H x V = Z x X = +Y (pad goes outward)

circles D70 at (h,v)=(0,0) and (260,0)

CATIA FEATURE RECIPE (Part Design)

1 Sketch.1 plane y=0 (H=+X,V=+Z): stadium - R50 at (0,0) and (0,260), lines x=+-50 -> Pad.1 Mirrored half 30

2 Sketch.2 plane y=30 (H=+Z,V=+X; normal +Y): 2x circle D70 at (0,0),(260,0) -> Pad.2 L=5 (+Y)

Local origin = J2 axis point; world translation in assembly = (0, 85, 330). J3 axis at local z=260.

Lower arm occupies world Y 55..115 (touches turret shoulder face y=55); caps are cosmetic.

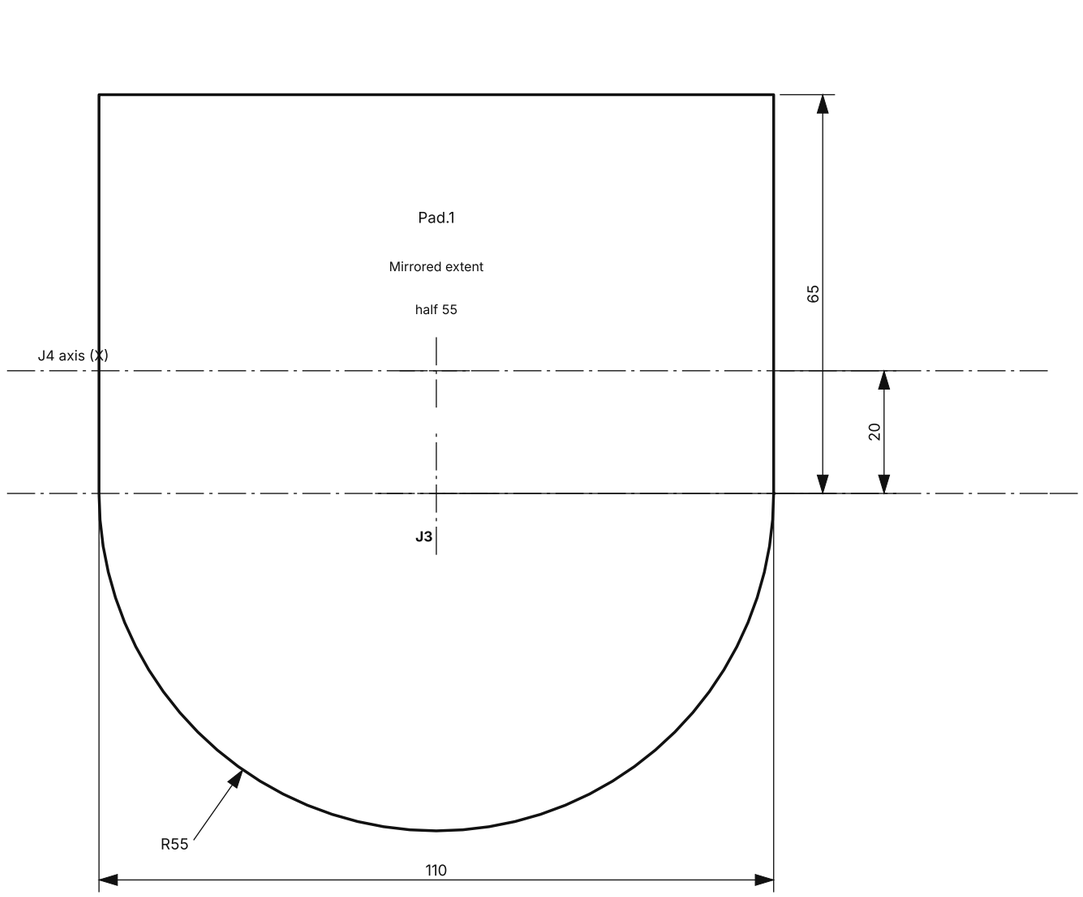
6-Axis Compact Robot Arm (LR Mate 200iD/4SC style) - approximate

P03_LowerArm - sketch profiles and extrusion

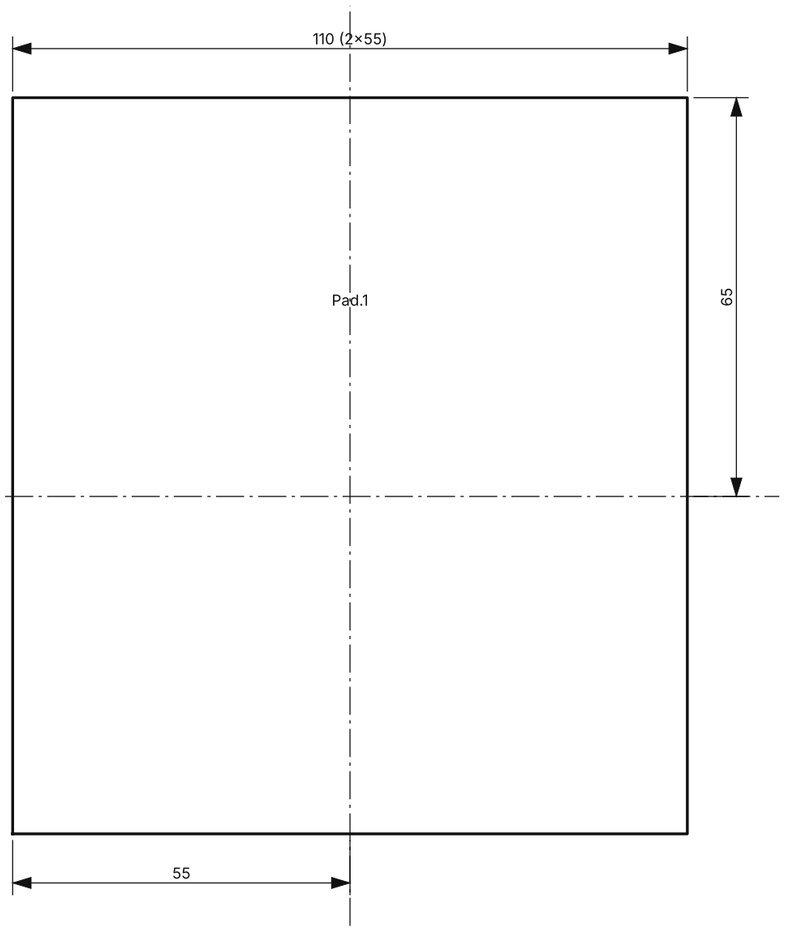
SCALE	UNITS	PROJECTION	SHEET	DATE
1:2	mm		1st angle 4/8	2026-11-11

Dimensions approximate - verify before manufacture

VIEW A - FRONT (XZ): Sketch.1



VIEW B - SIDE (YZ): thickness



CATIA FEATURE RECIPE (Part Design)

```
1 Sketch.1  plane y=0 (H=+X,V=+Z), closed profile:
     bottom semicircle R55 centre (0,0) from (-55,0) through (0,-55) to (55,0);
     (55,0)-(55,65)-(-55,65)-(-55,0)                -> Pad.1  Mirrored extent, half 55 (Y +-55)
Local origin = J3 axis (world (0,0,590)).  Front face x=55 carries the J4 axis at local z=20
(world z=610).  Forearm P05 is placed at (55,0,610).
```

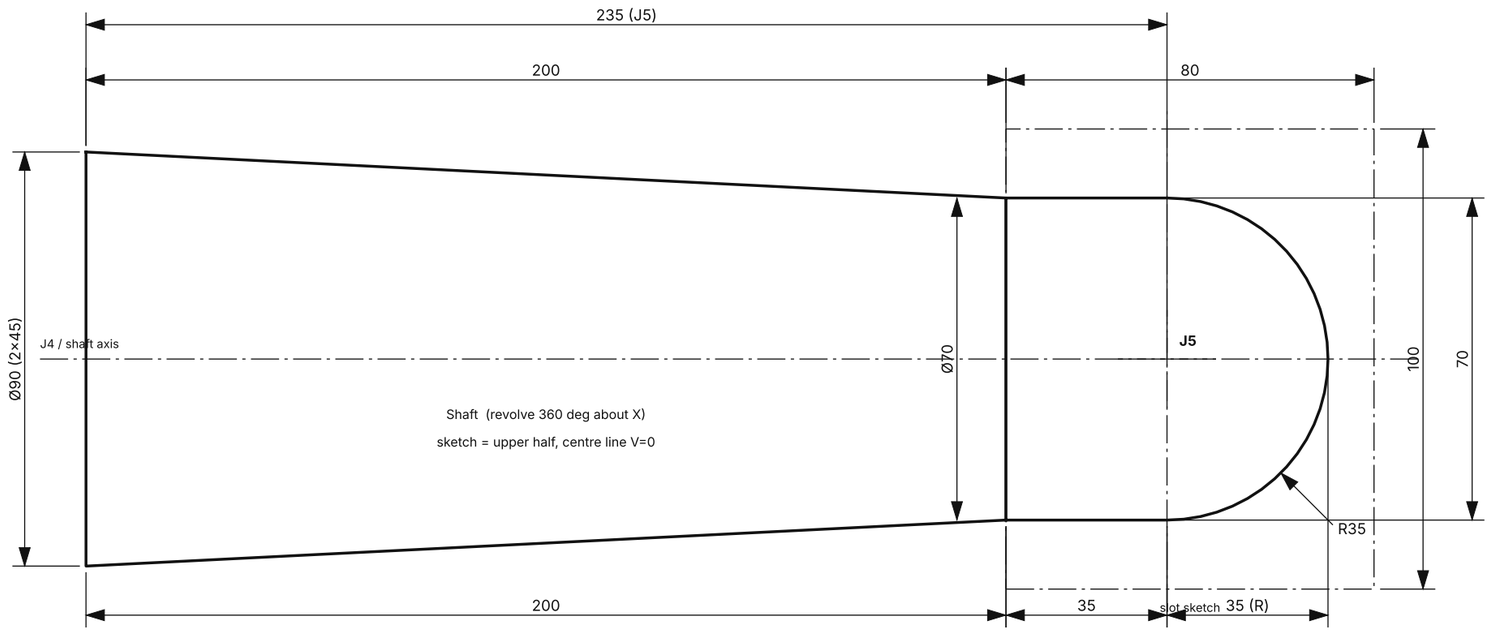
6-Axis Compact Robot Arm (LR Mate 200iD/4SC style) - approximate

P04_ElbowHousing - sketch profile and extrusion

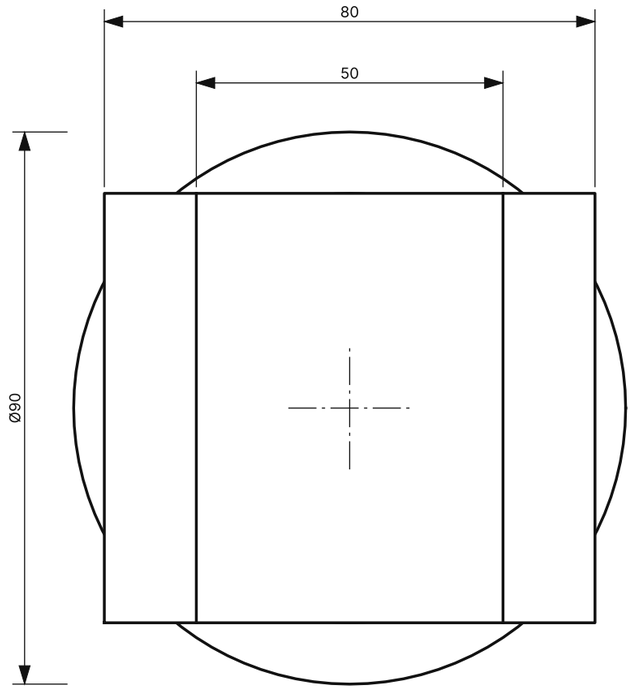
SCALE	UNITS	PROJECTION	SHEET	DATE
1:1	mm	1st angle	5/8	2026-11-11

Dimensions approximate - verify before manufacture

VIEW A - FRONT (XZ): Sketch.1 shaft profile (revolve), Sketch.2 yoke, Sketch.3 slot

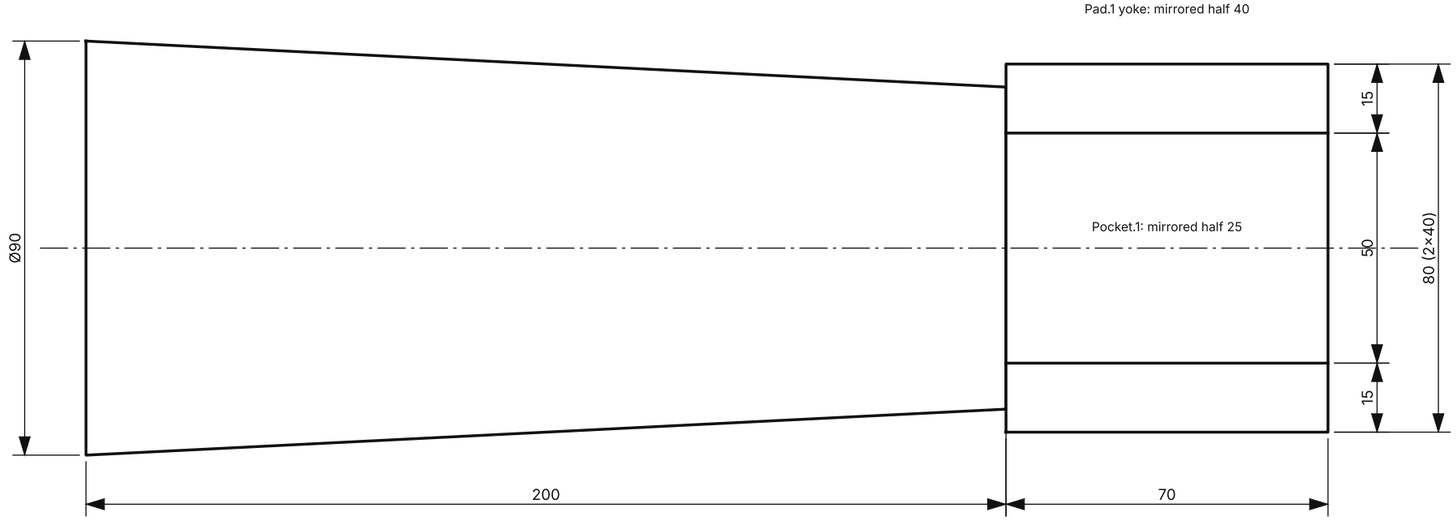


VIEW C - END (YZ) 1:1



yoke Z +-35 (70)

VIEW B - TOP (XY): extrusion thickness of yoke / slot



CATIA FEATURE RECIPE (Part Design)

```
1 Sketch.1  plane y=0 (H=+X,V=+Z): (0,0)-(0,45)-(200,35)-(200,0)-(0,0); construction line (0,0)-(200,0) = CenterLine
                                                  -> Shaft.1  360 deg about X  (cone D90 -> D70, x 0..200)
2 Sketch.2  plane y=0: (200,-35)-(235,-35), arc R35 c(235,0) to (235,35), (235,35)-(200,35), close -> Pad.1 mirrored half 40
3 Sketch.3  plane y=0: rect X 200..280, Z -50..50                        -> Pocket.1 mirrored half 25 (prongs 15 thick)
Local origin = J4 axis at elbow front face (world (55,0,610)).  J5 axis at local x=235 (world x=290).
```

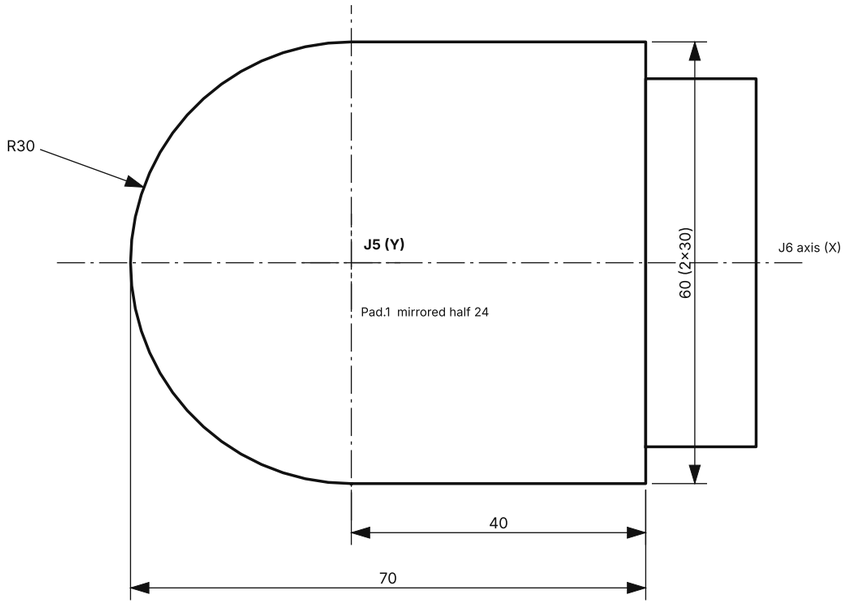
6-Axis Compact Robot Arm (LR Mate 200iD/4SC style) - approximate

P05_Forearm - sketch profiles and extrusion

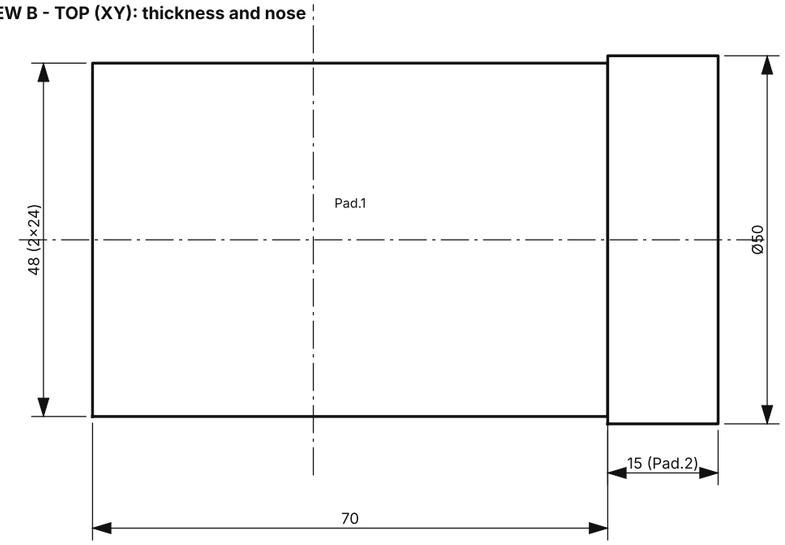
SCALE	UNITS	PROJECTION	SHEET	DATE
1:1.33	mm		1st angle 6/8	2026-11-11

Dimensions approximate - verify before manufacture

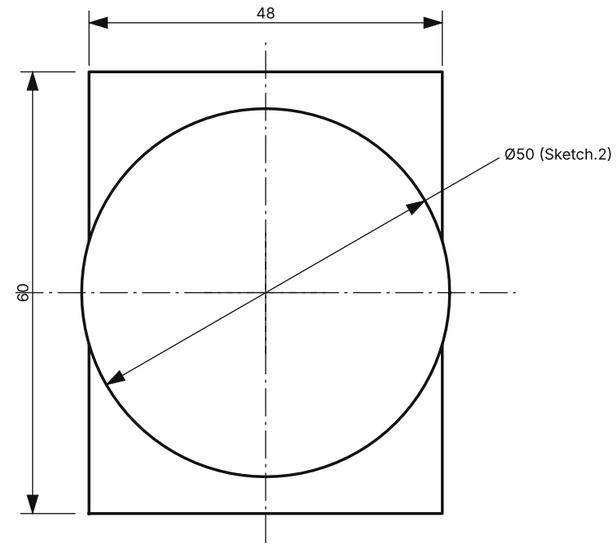
VIEW A - FRONT (XZ): Sketch.1 wrist profile



VIEW B - TOP (XY): thickness and nose



VIEW C - END (YZ): nose sketch



Sketch.2 on plane x=40:
H=+Y, V=+Z (normal +X)
circle D50 at (0,0)

CATIA FEATURE RECIPE (Part Design)

```
1 Sketch.1  plane y=0 (H=+X,V=+Z): (0,-30)-(40,-30)-(40,30)-(0,30), arc R30 c(0,0) through (-30,0) back to (0,-30)
                                                          -> Pad.1  Mirrored extent half 24 (Y +-24)
2 Sketch.2  plane x=40 (H=+Y,V=+Z, normal +X): circle D50 at (0,0)  -> Pad.2  L=15 (+X), x 40..55
Local origin = J5 axis (world (290,0,610)); wrist fits in the 50 mm forearm slot (24 < 25).  J6 axis along +X at z=0.
```

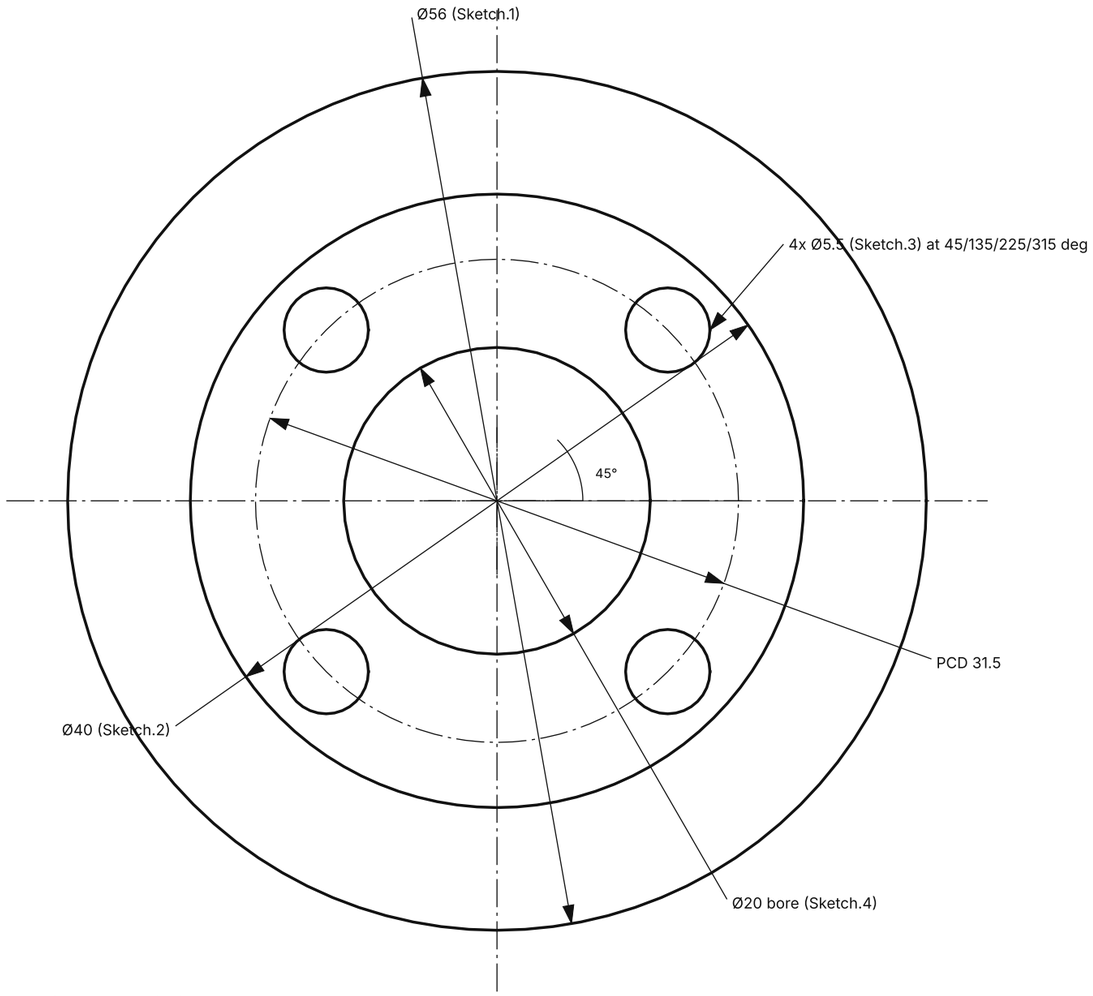
6-Axis Compact Robot Arm (LR Mate 200iD/4SC style) - approximate

P06_Wrist - sketch profiles and extrusion

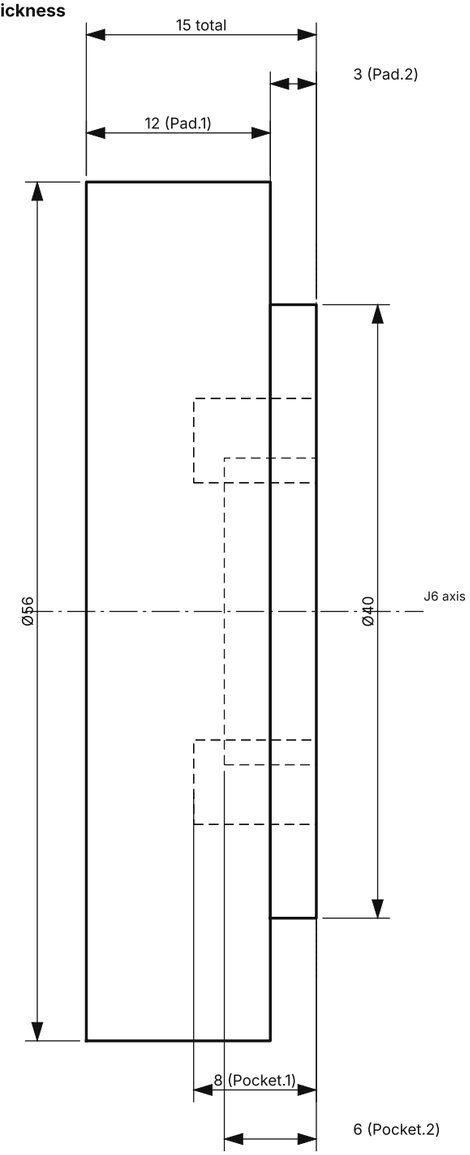
SCALE	UNITS	PROJECTION	SHEET	DATE
1.2:1	mm	1st angle	7/8	2026-11-11

Dimensions approximate - verify before manufacture

VIEW A - END VIEW (YZ), looking along -X: Sketch.1,.2 (disks), Sketch.3,.4 (pockets)



VIEW B - FRONT (XZ) 2.5:1: thickness



CATIA FEATURE RECIPE (Part Design)

```
1 Sketch.1  plane x=0 (H=+Y,V=+Z, normal +X): circle D56 at (0,0)               -> Pad.1  L=12 (+X)   x 0..12
2 Sketch.2  plane x=12 (H=+Y,V=+Z):          circle D40 at (0,0)               -> Pad.2  L=3  (+X)   x 12..15
3 Sketch.3  plane x=15 (H=+Y,V=+Z):          4x circle D5.5, PCD 31.5, 45/135/225/315
                                                                               -> Pocket.1 Mirrored extent half 8
4 Sketch.4  plane x=15 (H=+Y,V=+Z):          circle D20 at (0,0) (centre bore)  -> Pocket.2 Mirrored extent half 6
Hole centres (Y,Z): (+-11.137, +-11.137).  Local origin = J6 axis at x=55 from J5 (world (345,0,610)); TCP = (360,0,610).
```

Pockets are Mirrored extent about plane x=15:
bolt holes half 8 (x 7..15 effective), bore half 6 (x 9..15).
TCP face x=15; J5 face to TCP: 55 (nose) + 15 = 70 mm.

6-Axis Compact Robot Arm (LR Mate 200iD/4SC style) - approximate

P07_Flange - sketch profiles and extrusion

SCALE	UNITS	PROJECTION	SHEET	DATE
2.5:1	mm	1st angle	8/8	2026-11-11

Dimensions approximate - verify before manufacture